

Improving Dexterous Piano Playing Policies in PianoMime

Reward shaping, residual calibration, and generalist-policy imitation for robotic piano
Deep Reinforcement Learning Course Project 2026 | Dexterous Piano Track

Abstract & Motivation

Dexterous piano playing requires accurate note timing and smooth, physically plausible hand control. We study PianoMime-style policies under two settings: a single-song specialist and a multi-song generalist. Our final specialist method combines light smoothness regularization, onset-aware reward shaping, and residual-action calibration, improving Happy_8 F1 from 0.872 to 0.919. For the generalist policy, smooth diffusion objective fine-tuning gives the best mean F1 on five unseen clips.

- Motivation: reduce missed note onsets without making hand motion overly conservative.
- Integrity: we keep failed attempts, because they explain why the final method is light rather than heavily regularized.

RL Formulation

- State**
- proprioception, piano key state, future note targets, residual-policy context
- Action**
- low-level hand control or residual correction over IK-guided motion
- Reward**
- note precision/recall, mimic quality, timing, and smoothness terms
- Metric**
- F1 over played vs. target notes, plus qualitative videos
- Eval**
- Happy_8 specialist; five unseen clips for generalist policy

Method

- 1. Diagnose**
Ablate residual scale, extra PPO updates, and smoothness strength.
- 2. Shape Rewards**
Use light smoothness and onset-aware timing instead of heavy penalties.
- 3. Calibrate**
Tune residual correction magnitude to improve recall without extra presses.

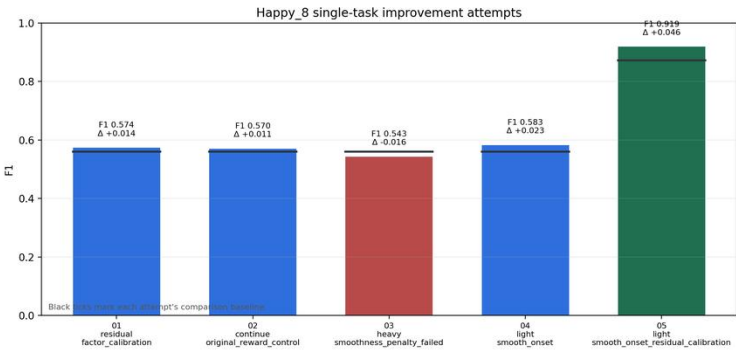
Specialist F1
0.9187

Specialist ΔF1
+0.0463

Best Generalist F1
0.6256

Generalist ΔF1
+0.0105

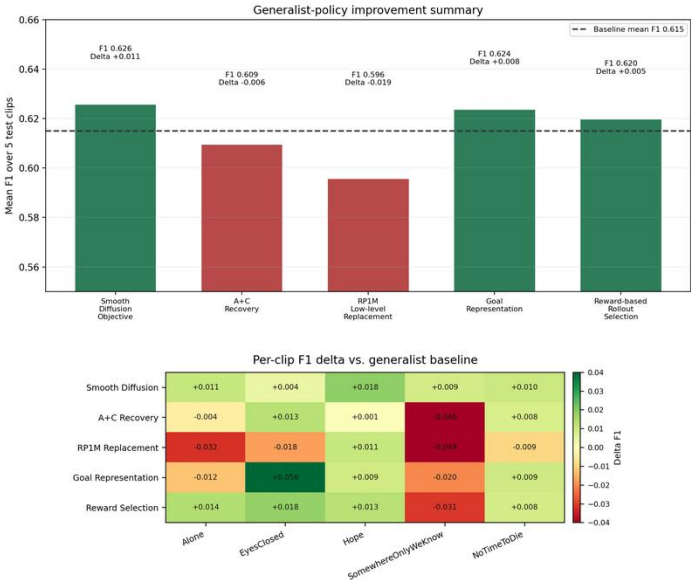
Single-task Improvement Attempts



Heavy smoothness fails by reducing recall. Extra PPO updates alone help only slightly. The final method keeps smoothness light and combines onset timing with residual calibration.

- Failed attempt: heavy smoothness lowers F1 to 0.543 by reducing recall.
- Positive intermediate: light smoothness + onset reward reaches F1 0.583.
- Final method: adding residual calibration reaches F1 0.919, Delta +0.046.

Generalist Results on Five Unseen Clips



Smooth diffusion objective is the strongest average improvement. Goal representation also improves the mean and is especially helpful on EyesClosed_1. RP1M replacement and A+C recovery are useful negative comparisons.

Future Directions

- Tune smoothness adaptively by tempo or local note density instead of using one global penalty.
- Combine goal-side-channel representations with diffusion objective fine-tuning for generalist policies.
- Add motion-quality metrics and synchronized comparison videos beyond note-level F1.

Note: single-task baseline package is pending; final specialist comparison uses released Happy_8 rollout baseline.